Daniil Privezentsev

AI Agent Developer | Multi-Agent Systems Research

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Research Interests

Reliable and resource-efficient language-model agents: multi-agent cooperation, adaptive agent granularity, verification and reflection, persistent memory and provenance, task-graph orchestration, and selective recovery. I investigate when separating agents improves verified engineering outcomes and when it adds unnecessary cost or correlated errors, accounting for tokens, latency and monetary cost. My proposed PhD direction combines an auditable orchestration controller, independent executable checks and provenance-aware recovery for coding and tool-use workflows.

Education

HSE University — *Moscow, Russia*
M.Sc., System and Software Engineering — 2025 - 2027 (Expected)

- Average score: **8.38/10**; GPA: **8.10/10**; expected completion: **10 July 2027**.
- Relevant study: formal methods, software architecture, applied systems analysis and software engineering methodology. Planned coursework: Multi-Agent Systems, Machine Learning and Secure Software Development.

MIREA - Russian Technological University — *Moscow, Russia*
B.Sc., Software Engineering — 2021 - 2025

- GPA: **4.8/5.0**; qualification in Software Engineering awarded 1 July 2025.
- Thesis: *Computer Application-Game for Developing Spelling Skills in Primary School Pupils*; final thesis grade: excellent.

Beijing Institute of Technology — *Beijing, China*
Exchange programme, Software Engineering; School of Computer Science and Technology. GPA: 87/100. — 2024 - 2025

Professional and Research Experience

VK — *Moscow, Russia*
Head of AI Agents Department — Jul. 2026 - Present

Sber — *Moscow, Russia*
AI Agent Development Team Lead — May 2026 - Jul. 2026

SberTech — *Moscow, Russia*
Senior AI Agent Developer — Mar. 2025 - May 2026

MIREA - Russian Technological University — *Moscow, Russia*
Researcher / Backend Developer, Department of Mathematical Support and Standardisation of Information Technologies (MOSIT) — 2022 - 2023

- Research and backend development for university and educational applications, including publication-request workflows.

MIREA - Russian Technological University — *Moscow, Russia*
Researcher / Backend Developer, Applied Mathematics Department — One-year appointment

Technical Skills

Programming	Python; Go (goroutines, channels, gRPC); C++ (STL, OpenGL, SFML); C#
AI Agents	LLM applications, RAG, LangGraph, MCP, prompt engineering
Systems & Tooling	Linux, Bash, Vim, GDB, strace, Ruff, pre-commit; static analysis with Svace
Software Development	Flask, Streamlit, Unity; configuration and development tooling
Languages	Russian (native); English (B2, self-assessed); Chinese (HSK 3)

Research Manuscripts

[1] **Hybrid-INoT: a hybrid agent architecture with internal role segregation for engineering tasks under long shared context.** (2026). Submitted to the FSE 2027 Research Track; under review, not accepted. Studies planner/worker/critic roles within one model call, coupled with external tools and verification. Token savings depend on the model and task; general quality non-inferiority has not been established. Research repository; replication package.

[2] **Swarm-INoT.** (2026). Adaptive roles, versioned task graphs and selective replay after failures. Evaluated on 72 held-out assignments in six project clusters; adaptive routing did not uniformly outperform fixed baselines. A paired corruption study examined recovery costs; large-scale simulations provide structural rather than end-to-end performance evidence.

[3] **Does Agent Decomposition Pay Under a Shared Budget? A Preregistered Response Surface on Function Calling.** (2026). A study of 720 intention-to-treat episodes: 20 held-out BFCL tasks, four hosted models, three agent topologies and three completion budgets. Current results show no demonstrated adaptive-policy gain and a high invalid-call rate. A separate validation run reached its spending ceiling and supports no scored quality claim.

[4] **Agent Granularity under Matched Token Budgets: A Preregistered Measurement Study with Small Open-Weight Code Models.** (2026). An earlier manuscript comparing four orchestration regimes under identical completion-token limits using five small open-weight code models. This study is retained as Study 0 in the later response-surface paper above.

Published Articles

[1] **Privezentsev, D. A.** (2024). Analysis of the IT Sphere in the Russian Economy and Searching for the Most Effective Measures of Government Support for Training New Personnel. In *Grani Nauki: International Research Competition proceedings*, pp. 10-13. Nauka i Prosveshchenie. [In Russian.] **First-place winner, Economics; First-Degree Diploma.** Full text.

[2] **Privezentsev, D. A.** (2024). Gaming Innovations in Education: Effectiveness of Learning and Development of Cognitive Abilities Through the Creation of Educational Computer Games. In *Best Scientific Research: proceedings of the VI International Scientific and Practical Conference*, pp. 24-26. Nauka i Prosveshchenie. [In Russian.] Full text.

Selected Projects

[1] **MIREA Chess** - lead developer of a 3D chess application in VR (C#, Unity, Photon). Russian software registration No. 2023661302.

[2] **PolyMAS Studio** - desktop application for configuring holonic multi-agent systems and studying inter-agent request routing: nested agent groups, communication graphs and execution-event inspection (Rust, Tauri, React, MCP, Codex CLI). Development prototype; private repository, materials available on request.

[3] **ABVGDeyka** - an educational spelling game that informed my undergraduate thesis and published research; recognised in two separate competitions in 2021 (see awards below).

[4] **MOSIT Admin** - a publication-request workflow application (Python, Flask, SQLite). Public project excerpt.

Awards and Additional Information

2024	First place, Economics; First-Degree Diploma	*Grani Nauki (international competition)*
2021	First place, Applied Programming, Mathematics and Information Technology	*Engineers of the Future*
2021	Winner, Cognitive Science and Informatics	*Moscow City Competition (MGK)*

PhD availability: autumn 2027, following MSc completion. **Interests:** Chinese language and culture, international academic exchange, travel and table tennis. I value reproducible experiments, explicit limitations and inspectable software artifacts.